

Household Service Management System

User Manual & Implementation Guide

1. Introduction

Welcome to the Household Service Management System! This platform connects customers needing household services (plumbing, electrical, cleaning, etc.) with verified service providers. The system is built with a modern MERN stack (MongoDB, Express, React, Node.js) and features a role-based architecture.

This manual will guide you through setting up the project and using it from the perspective of its three main roles: **Customer**, **Service Provider**, and **Admin**.

2. Installation & Setup

Prerequisites

- **Node.js** (v14 or higher)
- **MongoDB** (Running locally on default port 27017 or via MongoDB Atlas)

Steps to Run

1. **Start MongoDB:** Ensure your MongoDB server is running in the background.
2. **Start the Backend:**
 - Open a terminal and navigate to the `backend` folder.
 - Run `npm install` to install dependencies.
 - Configure the `.env` file with your `MONGO_URI` and `JWT_SECRET`.
 - Run `npm run dev` to start the backend server on port 5000.

3. Start the Frontend:

- Open a new terminal and navigate to the `frontend` folder.
- Run `npm install` to install dependencies.
- Run `npm run dev` to start the React development server.

4. Seed Initial Data:

- Open a terminal in the `backend` folder and run `node utils/seedCategories.js` to populate the database with default service categories and a default Admin user.

3. User Roles & Features

A. The Customer

Customers are everyday users looking to book a service.

1. **Registration:** Go to the `/register` page and select "Customer".
2. **Dashboard:** Upon login, customers land on their dedicated dashboard showing active/pending bookings and total spent.
3. **Browsing Services:** Customers can browse available service categories and view providers offering those services.
4. **Booking a Service:**
 - Click on a Provider's profile and click "Book Now".
 - Select the desired service, date, time, and provide an address/description.
 - The booking will initially be marked as **Pending**.
5. **Managing Bookings:** Customers can view their history and cancel a booking as long as it is still Pending or Accepted.

B. The Service Provider

Service providers are professionals offering their skills on the platform.

1. **Registration:** Register as a "Service Provider".

2. **Profile Setup:** Before accepting jobs, a provider must set up their profile. They can select which service categories they offer, set their experience level, and toggle their availability to "Available".
3. **Managing Requests:**
 - From the "Booking Requests" dashboard, providers will see all incoming requests.
 - Providers can **Accept** or **Reject** pending bookings.
 - Once accepted, a provider can mark the job as **In Progress**.
 - Upon completion, the provider clicks **Mark Complete** and enters the **Final Price** charged to the customer.

C. The Admin

Admins manage the overall platform data.

1. **Access:** Log in using admin credentials.
 2. **Categories:** Admins can create, edit, and delete service categories. (e.g., adding "AC Repair" or "Pest Control").
 3. **Oversight:** Admins have view-access to all bookings and users across the platform, ensuring quality and resolving disputes.
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4. Booking Lifecycle (State Transitions)

The system enforces strict rules on how a booking progresses:

- **Pending:** Created by the customer. Awaiting provider action.
 - **Accepted:** The provider agrees to the job.
 - **In Progress:** The provider has started the work.
 - **Completed:** The provider finishes the work and sets the final price.
 - **Cancelled:** The customer cancels the request (only allowed before the job is in progress).
 - **Rejected:** The provider declines the request.
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5. Technical Stack Overview

- **Database:** MongoDB & Mongoose (Schema validation, aggregation pipelines).
 - **Backend:** Node.js & Express.js (RESTful APIs, JWT Authentication, Role-based middleware).
 - **Frontend:** React (Vite), React Router v6, Context API, Axios, Tailwind CSS (Custom styling).
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Generated for the Household Service Management System project.